

10/02/2026: Midterm Review Part 1: Proofs

CSCI 246: Discrete Structures

October 2, 2026

Announcements

At beginning of class Monday, you can optionally retake any quiz of **your choice!**

The study guides will be the same. The questions will be different (in general).

Today's Agenda

- Quiz (20 mins)
- Review of proof techniques (5-10 mins)
- Group exercises (20-25 mins)

Review of proofs

Proof Template: Direct proof of an if-then (if A then B) theorem

1. Write down the if-then statement you're trying to prove.
2. At the beginning of the proof, write down the *antecedent* (the A in if A then B) as your assumption.
3. At the end of the proof, write down the *consequent* (the B in if A then B) as your conclusion.
4. Unravel the definitions, working forward from the beginning of the proof and backward from the end of the proof.
5. Forge a link between the two halves of the argument.

Example

Proposition. The square of an odd integer is odd.

Proof.

Annotation	Main Text
Convert Prop. to “if-then” form	We show that if x is an odd integer, then x^2 is odd.
State antecedent	Let x be an odd integer.
Unravel defs.	Then by definition of <i>odd</i> , there is an integer a such that $x = 2a + 1$.
*** The glue ***	So $x^2 = (2a + 1)(2a + 1) = 4a^2 + 4a + 1 = 2(2a^2 + 2a) + 1$.
Unravel defs.	So there is an integer b (where $b = 2a^2 + 2a$) such that $x^2 = 2b + 1$.
State consequent	Hence, x^2 is odd.

Proof Template: Direct proof of an if-and-only-if theorem

To prove a statement of the form “A iff B”:

- ($\implies$) Prove “If A, then B”.
- ($\impliedby$) Prove “If B, then A”.

Scheinerman Proof Template 2

Proof Template: Refuting a false if-then statement via a counterexample

To disprove a statement of the form “If A then B ”:
Find an instance where A is true but B is false.

Scheinerman Proof Template 3

Proof Template: Refuting a false if-then statement via a counterexample

To disprove a statement of the form “If A then B ”:

Find an instance where A is true but B is false.

Scheinerman Proof Template 3

Note. Often, you'll need to **justify** that A is true (by showing that the conditions of the relevant definition is satisfied.)

Proof Template: Proving one set is a subset of another

To show $A \subseteq B$:

Show that if $x \in A$, then $x \in B$.

Scheinerman Proof Template 6

Proof Template: Proving two sets are equal

Let A and B be sets. To show $A = B$:

- Show $A \subseteq B$.
- Show $B \subseteq A$.

Scheinerman Proof Template 6

Proof Template: Proving two sets are equal

Let A and B be sets. To show $A = B$:

- Show $A \subseteq B$.
- Show $B \subseteq A$.

Scheinerman Proof Template 6

Remark. In other words, we must show:

- If $x \in A$, then $x \in B$.
- If $x \in B$, then $x \in A$.

Group exercises

Ashley, Isabelle: 7	Hager, Nathan: 11	Putnam, John: 5
Bailey, Tanna: 7	Horner, Jayden: 7	Ramos, Vydel: 10
Blaisdell, Olyver: 5	Johnson, Jaret: 3	Rathbun, Jedidiah: 9
Brown, Ashton: 1	Kintzel, Samantha: 1	Roberts, Matthew: 13
Brown, Jonathan: 2	Lameres, Alexis: 9	Roller, Nathan: 4
Cardenas, Dominic: 2	Lee, Holden: 5	Ryan, Otis: 3
Cutler, George: 11	Mayala, Reece: 6	Shevtsov, Timothy: 1
Fabik, Roland: 8	McLean, Connor: 12	Sinclair, Aidan: 11
Fellows, Audrey: 10	Mcbride, Daniel: 10	Smith, Charles: 6
Fitzgerald, Ryan: 8	Miller, Maverick: 2	Stensland, Emma: 13
Fuller, Ryan: 6	Moss, Emily: 8	Thiessen, Logan: 9
Griffen, Clara: 4	Mousseau, Matthew: 3	Thompson, Jaiden: 4
Grutsch, Tanner: 12	Neuman, Evan: 13	Vath, Ethan: 12

Group exercises

1. Prove that the product of two even integers is even.
2. Let x be an integer. Prove that $0 \mid x$ if and only if $x = 0$.
3. Disprove: If p and q are prime, then $p + q$ is composite.
4. Let $A = \{x \in \mathbb{Z} : 4 \mid x\}$ and let $B = \{x \in \mathbb{Z} : 2 \mid x\}$. Show that $A \subseteq B$.
5. Let $A = \{n \in \mathbb{Z} : 3 \mid n\}$ and let $B = \{n \in \mathbb{Z} : 3 \mid (n + 6)\}$. Show that $A = B$.

Reference: Definitions

- Let a, b be integers. We say that a is **divisible** by b , written $b \mid a$, if there is an integer c such that $bc = a$. We call b a **divisor** of a .
- An integer is called **even** if it is divisible by 2. (That is, an integer x is even if there is an integer c such that $x = 2c$.)
- An integer p is called **prime** if $p > 1$ and the only positive divisors of p are 1 and p .
- A positive integer a is called **composite** if there is an integer b such that $1 < b < a$ and $b \mid a$.